

Differentiable functions – 3

Convexity and differential relations

Homework

1 сентября 2026 года

Theoretical minimum. Convexity through chords, tangents and the monotonicity of the derivative; regularity of convex functions; logarithmic convexity $f''f \geq (f')^2$; a C^2 function as a difference of convex ones; the operator $f \mapsto f + \lambda f'$ and the exponential multiplier; counting zeros of a differential expression through Rolle.

1. (IMC 1994) Let f have $n + 1$ derivatives at every point of $\mathbb{R}$. Show that for every pair $a < b$ with $\ln \frac{f(b)+f'(b)+\dots+f^{(n)}(b)}{f(a)+f'(a)+\dots+f^{(n)}(a)} = b - a$ there is $c \in (a, b)$ with $f^{(n+1)}(c) = f(c)$.
2. (MIPT 2006) Let $f \in C^2([0, 1])$. Prove that there are convex $g, h: [0, 1] \rightarrow \mathbb{R}$ with $f = g - h$.
3. (VJIMC 2012) Let $f: [0, 1] \rightarrow [0, 1]$ be differentiable with $|f'(x)| \neq 1$ for all x . Prove that there are unique $\alpha, \beta \in [0, 1]$ with $f(\alpha) = \alpha$ and $f(\beta) = 1 - \beta$.
4. (Putnam 1978) Let $0 < x_i < \pi$ for $i = 1, \dots, n$ and put $x = \frac{x_1 + \dots + x_n}{n}$. Prove that $\prod_{i=1}^n \frac{\sin x_i}{x_i} \leq \left(\frac{\sin x}{x}\right)^n$.
5. (Putnam 2015) Let f be three times differentiable on $\mathbb{R}$ with at least five distinct real zeros. Prove that $f + 6f' + 12f'' + 8f'''$ has at least two distinct real zeros.
6. (Putnam 2011) Let f and g be defined on an open interval containing 0, with g nonzero and continuous at 0. If fg and f/g are differentiable at 0, must f be differentiable at 0?

7. (Putnam 1988) A common mistake is to believe that $(fg)' = f'g'$. For $f(x) = e^{x^2}$, determine with proof whether there are an open interval (a, b) and a nonzero function g on it for which this false rule holds.
8. (Putnam 2023) For a positive integer n let $f_n(x) = \cos x \cos 2x \cos 3x \cdots \cos nx$. Find the smallest n with $|f_n''(0)| > 2023$.
9. (Putnam 1994) Find all real k with the property: for every positive differentiable f satisfying $f'(x) > f(x)$ for all x there is N with $f(x) > e^{kx}$ for all $x > N$.
10. (Putnam 2018) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be infinitely differentiable with $f(0) = 0$, $f(1) = 1$ and $f(x) \geq 0$ for all x . Show that there are a positive integer n and a real x with $f^{(n)}(x) < 0$.